THE VIRTUAL MEET EXPERIENCE

2017 - 2018

HS VIRTUAL CHALLENGE MEET #3

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2017-2018 Virtual Challenge Meet #3 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.7.25**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

What is $110101100111_2 - 4D9_{16}$?

A. 202032_4 B. 100010001110_2 C. 4216_8 D. $88E_{16}$ E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. 11 B. 12

C. 11.8 D. 0

E. There is no output due to an error.

```
int y=7/2;
y+=8.3;
out.print(y);
```

QUESTION 3

What is output by the code to the right(e's indicate blank spaces)?

A. its fun 002.34 B. itsfun00.234

C. its fun 002.33 D. its %s %06.2f

E. There is no output due to an error.

```
String s="fun";
double k=2.3351;
out.printf("its %s %06.2f",s,k);
```

QUESTION 4

What is output by the code to the right?

A. a B. r

C. e D. (empty space)

E. There is no output due to a syntax error.

```
String h="you are fake news";
h.substring(2,8);
out.print(h.charAt(5));
```

QUESTION 5

What is output by the code to the right?

A. truetrue B. falsetrue

C. falsefalse D. truetrue

E. There is no output due to a syntax error.

```
boolean t=false;
boolean f=true;
out.print(f&(!t^f|t));
out.print(f^(f&(t|f)));
```

QUESTION 6

What is output by the code to the right?

A. -8 B. -9

C. -8.0 D. -9.0

E. There is no output due to an error.

```
out.print(Math.floor(-8.5));
```

QUESTION 7

What is the output by the code to the right?

A. 128 B. 18

C. 19 D. 129

E. 9

```
int y=8,x=10;
if(y<x | ++y>x)
 out.print(1);
if(x-y == x/y)
 out.print(2);
out.print(y);
```

QUESTION 8

What is the output by the code to the right?

A. 12 B. 11

C. 10 D. 13

E. 9

```
int y=1,x=0;
while(y<1000){
 y*=2;x++;
}
out.println(x);
```

QUESTION 9

What is the output by the code to the right ?

A. 360 B. 678

C. 1368 D. 696

E. There is no output due to a syntax error.

```
String j="***";
for(int t=0;t<9;t++)
 j+="**";
for(int i=0;i<5;i++)
 j+=j;
for(int k=0;k<6;k++)
 j+="****";
out.print(j.length());
```

QUESTION 10

What is the output by the code to the right ?

A. 62 B. 1

C. There is no output due to a syntax error

D. There is no output due to a runtime error

E. Output cannot be determined until runtime.

```
int[][]mat=new int[][]
      {{2,1,2},{12,2,62},{70,8,10}};

out.print(mat[1][2]);
```

QUESTION 11

Which of the following could read in the following number for calculation purposes?

6

A. file.nextInt() B. file.nextDouble()

C. file.nextLine()

D. More than one of the above.

E. None of the above.

QUESTION 12

What is the output by the code to the right ?

A. 0 B. 1024

C. 1365 D. 212

E. There is no output due to a syntax error.

```
int y=1;
for(;y<1000;y++)
 y*=4;
out.print(y);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. III, IV, II, I B. IV, III, I, II

C. I, II, III, IV D. II, IV, III, I

E. IV, III, II, I

```
I. |
II. ==
III. >>
IV. instanceof
```

QUESTION 14

What is the output by the code to the right ?

A. 4 B. 64

C. 8 D. 32

E. 16

```
out.print(Short.SIZE);
```

QUESTION 15

What could replace **<1*>** in the code to the right?

A. <>
B. <Integer>
C. <String>
D. <E>
E. More than one of the above could replace **<1*>**

```
ArrayList<Object> a;
a=new ArrayList<1*>();
```

QUESTION 16

What is the output by the code to the right ?

A. 0
B. 34
C. 31
D. 42
E. There is no output due to a syntax error.

```
out.print(42^73&54);
```

QUESTION 17

What is the output by the code to the right ?

A. 550
B. 825
C. 450
D. 675
E. There is no output due to an error.

```
int total=0;
for(int y=0;y<10;y++)
 for(int x=0;x<y;x++)
    for(int v=0;v<5;v++)
      total+=v;
out.print(total);
```

QUESTION 18

What is output by the line marked //q18 code to the right?

A. true
B. false
C. 212
D. There is no output due to a runtime error.
E. There is no output due to a syntax error.

QUESTION 19

What is output by the line marked //q19 code to the right?

A. [000, 211, 343, 212, 555]
B. [000, 211, 212, 343, 555]
C. [000, 211, 555, 343, 212]
D. There is no output due to a runtime error.
E. There is no output due to a syntax error.

```
HashSet<String>h;
h=new HashSet<String>();
h.add("212");
h.add("000");
h.add("211");
h.add("555");
h.add("343");

out.print(h.add("212")); //q18

out.print(h);            //q19
```

QUESTION 20

What is the big O runtime of adding a value to a stack?

A. O(n)
B. O(n^2)
C. O(n !)
D. O(1)
E. None of the above.

QUESTION 21

Which sorting algorith is implemented in the code to the right?

A. Selection sort

B. Insertion sort

C. Quick sort

D. Merge sort

E. Spaghetti sort

```
int[] k={2, 4, 3, 5, 212, 212, 9};
ArrayList<Integer> a;
a=new ArrayList<Integer>();
for(int y:k)
  a.add(y);
Collections.sort(a);
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right, so that the instance variable a would be set to the value of the parameter a?

A. `super.`

B. `this.`

C. `self.`

D. A and B

E. A and C

QUESTION 23

Assuming that **<1*>** is filled correctly, what is the output by the line marked `//1` in the client code to the right?

A. `12 7`

B. `7 7`

C. `12 12`

D. `7 12`

E. There is no output due to an error.

QUESTION 24

Assuming that **<1*>** is filled correctly, what is the output by the lines marked `//2` in the client code to the right?

A. `2 4 32`

B. `4 32`

C. `8 32`

D. `8 16`

E. This is no output due to an error.

QUESTION 25

Assuming that **<1*>** is filled correctly, what is the output by the line marked `//3` in the client code to the right?

A. `0`

B. `1`

C. `2`

D. Output cannot be determined until runtime

E. This is no output due to an error.

```
class A{
  private int a;
  public A(int z){
     a=z;
  }
  public A(){
     out.print(2+" ");
  }
  public int add(int k){
     return k+a;
  }
  public int mult(int k){
     return k*a;
  }
}

class B extends A{
  private int a,b;
  public B(int a){
     <1*>a=a;
     b=4;
  }
  public int mult(int h){
     return(h+a)*b;
  }
  public int fun(int x){
     return super.mult(x)/add(x);
  }
}
/////////////client code////////////
A a=new A(3);
out.println
        (a.add(4)+" "+a.mult(4)); //1
a=new B(4);                       //2
out.println
        (a.add(4)+" "+a.mult(4)); //2
out.println(a.fun(7)); //3
```

QUESTION 26

What is output by the code to the right?

A. `[1, 2, 3, 4, 5, 6]`

B. `[0, 0, 0, 0, 0, 0]`

C. `[212, 212, 212, 212, 212, 212]`

D. There is no output due to an error.

E. None of the above.

```
int[] z={1,2,3,4,5,6};
for(int y:z)
  y=212;
out.print(Arrays.toString(z));
```

QUESTION 27

Which of the following is not a valid java identifier?

A. `C00l`

B. `____`

C. `cooli0`

D. `212time`

E. All of the above are valid.

QUESTION 28

What is output by the line marked `//q28`?

A. `51`

B. `33`

C. `52`

D. `34`

E. `0`

QUESTION 29

What is output by the line marked `//q29`?

A. `0`

B. `6`

C. `3`

D. `-2`

E. `9`

QUESTION 30

What is output by the line marked `//q30`?

A. `165`

B. `166`

C. `243`

D. `244`

E. `0`

```
public static int recur(int s){
 if(s<0)
    return 3;
 return 3+recur(s-1)+recur(s-3);
}


//////client code/////////////////
out.println(recur(4)); //q28
out.println(recur(-2)); //q29
out.println(recur(8)); //q30
```

QUESTION 31

What could replace **<1*>** in the code to the right so that method `isEmpty` works as intended?

A. `size==0`

B. `first.next==null`

C. `size !=0`

D. A and B

E. B and C

QUESTION 32

What is the output by the following client code?

```
Node n1=new Node(212);
Node n2=new Node("A");
Node n3=new Node(17);
n1.next=n2;
n2.next=n3;
n3.next=n1;
Node n=n2;
for(int y=0;y<45;y++)
  n=n.next;
out.println(n.val);
```

A. `212`

B. `A`

C. `17`

D. There is no output due to an error.

E. Output cannot be determined until runtime.

QUESTION 33

What is output by the client code to the right?

A. `32`

B. `212`

C. `ABC`

D. There is no output due to an error.

E. Output cannot be determined until runtime.

QUESTION 34

Which data structure is implemented in the code to the right?

A. Queue

B. Array based list

C. Stack

D. Matrix

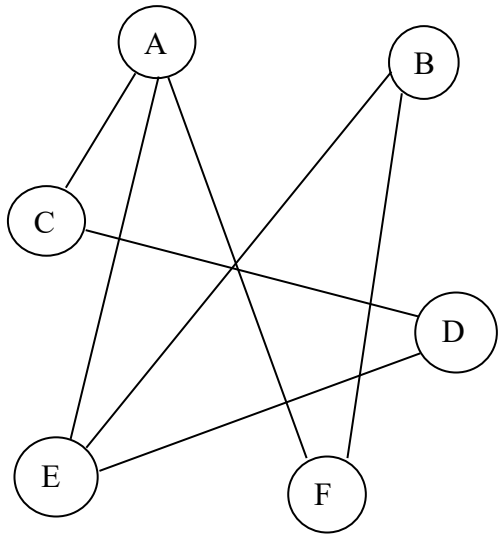
E. Hashtable

```
class Node {
  public Node next;
  public Object val;

  public Node( Object v ) {
          val = v;
  }
}

class Structure{
  Node first;
  Node last;
  int size;
  public Structure(){
          first=new Node(null);
          last=new Node(null);
          size=0;
  }
  public void add(Object o){
          last.next=new Node(o);
          last=last.next;
          size++;
          if(size==1)first=last;
  }
  public Object remove(){
          Node n=first;
          if(!isEmpty()){
            first=first.next;
            size--;
          }
          return n;
  }
  public Object look(){
          return first;
  }
  public boolean isEmpty(){
          return <1*>;
  }
  public Object get(int y){
          Node n=first;
          for(int h=0;h<y;h++)
            if(n!=null)n=n.next;
          return n.val;
  }
}


/////client code//////////////////
Structure s=new Structure();
s.add("212");
s.add(32);
s.add(new Node(45));
s.add("ABC");
s.add(null);
s.add(true);
s.remove();
out.println(s.get(1));
```

QUESTION 35

What kind of graph is represented to the right ?

A. undirected unweighted
B. connected weighted directed
C. weighted undirected
D. unweighted directed
E. connected weighted undirected



QUESTION 36

What is the most simplified path from node A to F ?

A. `AF`
B. `AEBF`
C. `ACDEBF`
D. `AEDCAF`
E. There is no such path between these two nodes.

QUESTION 37

What is output by the code to the right?

A. `1360`
B. `1507`
C. `1488`
D. `1469`
E. This is no output due to a syntax error.

```
String k="D0 yOu kNoW tH3 Way";
int tot=0;
for(char c:k.toCharArray())
  tot+=c;
out.print(tot);
```

QUESTION 38

What is the big O runtime of removing a value from a stack?

QUESTION 39

What is the value of the following prefix expression?

`+ * - / 6 7 + 4 5 * 9 3 * 2 1`

QUESTION 40

What is the pre-order traversal of the binary tree created when the following values are inserted in order ?

`7 212 17 4 2 91 3 2 17 9 -2`

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2017-2018 Virtual Challenge Meet #3
Computer Science - KEY

1) E
2) A
3) A
4) B
5) C
6) D
7) D
8) C
9) D
10) A

11) D
12) C
13) A
14) E
15) A
16) D
17) C
18) B
19) A
20) D

21) D
22) B
23) D
24) A
25) E
26) A
27) D
28) A
29) C
30) C

31) A
32) B
33) E
34) A
35) A
36) A
37) C
38) O(1)
39) -241
40) 7 4 2 2 -2 3 212 17 17 9 91

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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